

Lum Wen-Shen

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SUMMARY

Software Engineering student at Monash University building production AI agents and infrastructure for SME digital transformation. Shipped autonomous agentic systems, multi-modal RAG pipelines, and containerized deployments guiding 3–4 clients through Industry 4.0 adoption. Seeking applied research or engineering roles at the intersection of industrial AI, IIoT, and smart manufacturing.

EDUCATION

Monash University Malaysia

Bachelor of Engineering (Software Engineering)

Awards: *Monash High Achiever Award*

Oct 2024 – Present

CGPA : 3.725

EXPERIENCE

Part-time AI Agent Engineer | *Beyond Photography*

Feb 2026 – Present

- Architected modular agentic AI platform with tool-use orchestration and autonomous task execution, onboarded across 3–4 enterprise client deployments.
- Administered Proxmox VE cluster of 30–50 VMs, maintaining 10 concurrent production environments for live AI service deployments.
- Developing proof-of-concept AI automation systems for SME clients, enabling structured Industry 4.0 adoption by eliminating manual operational bottlenecks before full deployment.

AI Researcher Intern | *Beyond Photography*

Oct 2025 – Feb 2026

- Reduced deployment time from **20 minutes to 30 seconds**, saving **10 engineer-hours/month** across 30+ deployments.
- Developed LLM-powered WhatsApp chatbot for client-facing customer support and sales, automating inbound enquiry handling and lead qualification via structured conversation flows.
- Architected multi-modal **RAG pipeline** (PDF, image, voice) achieving **~90% retrieval accuracy** versus full-context baseline, with dynamic re-indexing via text extraction for scalable document updates.

Head Treasurer | *Monash University Engineering Club*

Jul 2025 – Present

- Automated financial synchronization via Python/Asana/Google Sheets APIs, saving **5 hours/week**.
- Implemented OAuth2-secured budget logic, achieving **100%** expenditure tracking accuracy.
- Oversaw **RM 300K+** annual budget across all Engineering Club allocations, processing ~5 claims and up to 10 purchase approvals weekly with **zero financial discrepancies**.

Co-working Space Coordinator | *HeadSpace SS15*

Jan 2024 – Present

- Maintained day-to-day operations and tenant relations for a co-working space serving **15–20 active accounts**, resolving facilities and administrative requests end-to-end.
- Organised **5+ corporate events** and community programs, managing vendor coordination, scheduling, and on-site execution.

PROJECTS

Budget Claim Analysis Dashboard

Jan 2026 – Present

- Built Python ETL pipeline and vanilla JS frontend delivering real-time operational analytics and automated reporting for multi-club financial oversight.
- Reduced reporting time by **80%** via automated data ingestion, transformation, and spending categorization.

AI Trading Agent | *Beyond Photography*

- Built autonomous sentiment-driven trading agent with structured tool calling and multi-step reasoning.

- Achieved 70% accuracy on directional market calls via prompt-driven decision framework.

Automated Door Control System | *Arduino, PIR Sensor, Servo Motors*

Coursework

- Designed motion-sensing entry system using PIR sensor to detect movement and trigger dual servo motors.
- Implemented sensor-to-actuator control loop for real-time mechanical response in embedded environment.

Intelligent Traffic Light Controller | *Arduino*

Coursework

- Built programmable traffic light system with custom state-machine logic for intersection flow management.
- Programmed timing sequences and safety interlocks in embedded C for Arduino microcontroller.

SKILLS & LANGUAGES

- **AI & Automation:** Agentic AI, RAG Pipelines, IIoT Integration, LLM Integration, n8n
- **Programming:** Python, JavaScript (Node.js), SQL, HTML/CSS
- **Infrastructure:** Proxmox VE, Docker, Nginx, Container Management, Git, REST APIs
- **Data & Analytics:** ETL Pipelines, Real-time Dashboards, Automated Reporting, Data Visualization
- **Languages:** English (Fluent), Malay (Native), Mandarin (Conversational), German (B1)